

Multiple Choice (10 marks)

1. Calculate the volume of 0.3 N base necessary to neutralize 3 liters of 0.01 N nitric acid.

- (a) 0.3 liters
- (b) 2 liters
- (c) 0.2 liters
- (d) 3 liters
- (e) 0.03 liters

2. How many milliliters of 0.250 M KOH does it take to neutralize completely 50.0 mL of 0.150 M H_3PO_4 ?

- (a) 75.0 mL
- (b) 90.0 mL
- (c) 60.0 mL
- (d) 120 mL
- (e) 30.0 mL

3. Compute the missing mass of $(^{19}_9\text{F})$ (*atomicweight* = 18.9984) *when the masses of proton, neutron and*

- 4. 0.1645 amu
- 5. 0.1987 amu
- 6. 0.1312 amu
- 7. 0.1588 amu
- 8. 0.1756 amu

4 liters of octane gasoline weigh 3.19 kg. Calculate what volume of air is required for its complete combustion at S.T.P.

- (a) 22,102.28 liters air
- (b) 58,650.00 liters air
- (c) 49,350.25 liters air
- (d) 65,292.15 liters air
- (e) 39,175.40 liters air

Without referring to a table, place the following hydro-carbons in order of increasing boiling points. (a) methane (d) neopentane (b) n-hexane (e) 2,3-dimethylbutane (c) n-undecane

- (a) 2,3-dimethylbutane < neopentane < n-hexane < n-undecane < methane
- (b) methane < 2,3-dimethylbutane < neopentane < n-hexane < n-undecane
- (c) n-undecane < n-hexane < neopentane < 2,3-dimethylbutane < methane
- (d) neopentane < 2,3-dimethylbutane < methane < n-undecane < n-hexane
- (e) 2,3-dimethylbutane < methane < n-hexane < neopentane < n-undecane

True / False (10 marks)

Answer True or False

- 6. True or False: The entropy change for the vaporization of water is $16.2 \text{ J mol}^{-1} \text{ K}^{-1}$, reflecting the energy distribution during the phase change.
- 7. True or False: The maximum wavelength of the earth's blackbody radiation at a mean surface temperature of 288 K is $2.5 \times 10^{\{-5\}} \text{ m}$.
- 7. True or False: The ammonium ion (NH_4^+) has an octet of electrons around the central nitrogen atom.
- 8. True or False: Hydrogen diffuses 4 times faster than oxygen gas at standard conditions due to its lower molar mass.
- 9. True or False: The molarity of the H_2SO_4 solution is 0.265 M , as determined by the neutralization reaction with NaOH .

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. A chemist discovered a new compound and observed a freezing point depression of water to -1.22°C upon adding 2.00 g of the compound. Discuss the methods used to calculate the molecular weight.
- 11. Calculate the number of molecules in $1.1 \times 10^8 \text{ kg}$ of phosgene gas used during World War I. Discuss the significance of this calculation in the context of chemical warfare.

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